Date: 2026-07-07
Request ID: 100048743

Hotplate		
Analyte(s)	Co, K, Pt	
SOP#		
Cocktail	15 mL HCl, 10 mL H2SO4, 5 mL HNO3	
sample(s)	**weight (g)**	**volume (mL)**
200128520_1	0.5099	50
200128520_2	0.5083	50

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LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128520	17.2239	19.2247	19.1458	3.94	0.9606

Co ICP analysis

sample	Dilution	conc. [mg/L]	Co_ppm [dried]	%Co [dried]
200128520_1	100/0.12	2.120	180348.49	18.03
200128520_2	100/0.12	2.212	188767.26	18.88

co result: 184557.87 ppm
co result: 18.46 %
co lot: *Manufacturer: Assurance, 29-22COY exp: 2027-06-30*

K ICP analysis

sample	Dilution	conc. [mg/L]	K_ppm [dried]	%K [dried]
200128520_1	250/0.7	0.605	22039.22	2.20
200128520_2	250/0.7	0.597	21834.29	2.18

k result: 21936.75 ppm
k result: 2.19 %
k lot: *Manufacturer: Assurance, 28-6KY exp: 2026-08-30*

Pt ICP analysis

sample	Dilution	conc. [mg/L]	Pt_ppm [dried]	%Pt [dried]
200128520_1	10/7	2.376	346.50	0.03
200128520_2	10/7	2.206	322.72	0.03

pt result: 334.61 ppm
pt result: 0.03 %
pt lot: *Manufacturer: Assurance, 28-2PTY exp: 2027-02-28*

Note(s): *Sample contained clumps/chunks. Ran as-is per Udari's instruction.*

ppm M+ = [conc.][volume][dilution]/[weight]

%M+ = [conc.][volume][dilution]/([weight]*10,000)

%MO = [oxide factor]*%M+

A = crucible
B = crucible + sample
C = crucible + sample after drying

%LOI = ([B-A]-[C-A])/(B-A)*100%

ppm M+ [dried] = ppm M+/([1-(%LOI)/100])

%M+ [dried] = %M+/([1-(%LOI)/100])

%MO [dried] = %MO/([1-(%LOI)/100])